Zuri Ye

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EDUCATION

CARNEGIE MELLON UNIVERSITY — Cumulative GPA: 3.91/4.0

Major in Electrical and Computer Engineering, Additional Major in Robotics — Class of 2029

Relevant Coursework: Computer Systems; Computer Vision; Data Structures & C; Analog Circuit Design; Digital Logic and Hardware Description Languages; Linear Algebra; Discrete; Calc III

TECHNICAL EXPERIENCE

RESILIENT INTELLIGENT SYSTEMS LAB (RISLAB) | GITHUB — November 2025 - Present

Undergraduate Researcher

- Researching real-time SLAM optimization for high-speed UAVs to mitigate motion blur and large inter-frame displacement
- Analyzing and implementing Gaussian Splatting-based monocular SLAM and relevant methodologies (visual-inertial odometry, optical flow, IMU preintegration, sensor fusion, loop closure, pose estimation, Lie Group Transformations) found in VINGS-Mono to leverage 3D mapping and SLAM pipelines
- Optimizing pipeline execution speed and investigating ML methodology to improve feature association across wide spatial gaps through application of machine learning techniques (factor graphs, GTSAM, GRUs, CNNs) in PyTorch and CUDA

TARTAN AUV — September 2025 - Present

Electrical Subteam Member

- Designing custom PCBs in Altium Designer (e.g. power distribution, servo breakout)
- Assembling PCBs and performing hands-on SMD, through-hole soldering, reflow soldering, and testing
- Executing general electrical tasks to integrate hardware (e.g. tether)
- Competed at RoboSub competition with team's submarine, capable of autonomously aiming and shooting torpedoes at targets, identifying symbols, and navigating around obstacles

HONEY HAVEN | GITHUB | GAME — January 2026 - May 2026

Project Co-Lead, Programming Lead, Art Lead

- Led full-lifecycle development and Itch.io release of a visual novel, winning 1st Place at a game expo with 200+ plays
- Architected modular GDScript systems to manage core game state, UI themes, and minigame integration
- Engineered a Twine-to-JSON pipeline dynamically parsing writer scripts into UI elements, character states, and scene cues
- Managed minigame and art subteams while producing original cover art, backgrounds, and cross-team workflows

USS HORNET MUSEUM — July 2023 - August 2025

Aircraft Restoration Group Volunteer

- Restored Vietnam aircraft (T-28B Trojan, S-3 Viking, F-4 Phantom II) with corrosion treatment, repainting, and power tools
- Maintained other exhibits, repairing tug engines, restoring drop tanks, and creating interactive museum activities

MIT LL BEAVER WORKS SUMMER INSTITUTE | GITHUB — July 2024 - August 2024

Unmanned Air Vehicles Student and Software Communications Lead

- Engineered IMMANUEL, a ROS2-based autonomous quadcopter, in a team based environment for task-driven competition
- Led node communication for controls and CV tasks (e.g. open loop control, obstacle avoidance) in Python and OpenCV

PERSONAL PROJECTS

HUAWEI U2800A CANDYBAR PHONE-IPOD — In Progress

- Refurbishing a Huawei u2800a cellular shell into a custom PCB MP3-player featuring Wi-Fi playlist sync, microSD offline storage, an embedded TFT display interface, USB-C LiPo power architecture, soft shutdown and software reset in KiCAD
- Designed around ESP32-WROOM-32E MCU, integrating SPI, I2C, and I2S, with plans for a partner Spotify-syncing app

SIR JESTER | GITHUB | GAME — February 2026

- Developed gameplay software for a lightweight, 2D retro-style game in 24 hours for a team hackathon
- Achieved size of 14kB zipped, using a single-file HTML, CSS, and JavaScript architecture, 8-bit music, and pixel art

ADDITIONAL SKILLS

Languages & Tools: C, Python, Java, R, MATLAB, OpenCV, ROS2, Linux, Bash, Git, GitHub
CAD & Fabrication: Altium Designer, KiCAD, OnShape, PCB assembly & soldering, 3D printing, microcontrollers
Technical Domains: computer vision, sensor fusion, embedded systems, machine learning